Test Report No. 7191276799-MEC22-ES
dated 8 Apr 2022
5585034

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SUBJECT:

Testing of sealant submitted by Bostik Findley (M) Sdn Bhd

TESTED FOR:

Bostik Findley (M) Sdn Bhd
Lot 112 & 113, Kawasan Perindustrian Senawang
70450 Seremban, Negeri Sembilan
Malaysia

Attn: Mr. Herman Chen / Mr. Wang SM

SAMPLE DESCRIPTION:

The following items were received on 10 Jan 2022 as shown:

Sample	Size	Quantity
'Bostik A330 Fill A Gap Flex Acrylic Sealant For Sealing And Gap Filling' (refer to Photo 1)	300ml / cartridge	12 cartridges

Photo 1 : 'Bostik A330 Fill A Gap Flex Acrylic Sealant For Sealing And Gap Filling'

PSB Singapore

TEST METHODS:

HDB Specifications Sealant - Semi-Precast Construction

Staining And Colour Change, UV Exposure

1. Adopted ASTM C510 : 2016 Standard Test Method For Staining And Colour Change Of Single Or Multi-Component Joint Sealants

Test equipment	:	QUV Weatherometer
Lamp designation	:	Fluorescent UVA 340 mm
Test cycle	:	8 hours UV exposure at 60±3°C and 4 hours condensation at 50±3°C, irradiance 0.89 W/m^2.nm (ASTM G154)
Exposure duration	:	100 hours
No. of determinations	:	4 samples: 2 samples with sealant and 2 samples without sealant (For UV Exposure) 2 control samples: 1 sample with sealant and 1 sample without sealant (Standard Conditions)

Staining And Colour Change, Standard Conditions In Distilled Water

Test apparatus	:	Container with distilled water
Test condition	:	Distilled water immersion for 1 minute, once a day, (5 days per week)
Test duration	:	14 days
No. of determinations	:	2 samples: 1 sample with sealant and 1 sample without sealant (For distilled water immersion) 2 control samples: 1 sample with sealant and 1 sample without sealant (Standard Conditions)

Extrudability

2. Adopted ASTM C1183/C1183M : 2013 (2018) Standard Test Method For Extrusion Rate Of Elastomeric Sealants

Test pressure	:	40 psi
No. of determination	:	1

Flow Properties

3. ASTM C639 : 2015 Standard Test Method For Rheological (Flow) Properties Of Elastomeric Sealants

Method	:	Test method for 'Type II' sealant
Test conditions	:	a) 4.4°C in environmental chamber for 4 hours b) 50°C in oven for 4 hours
No. of determinations	:	2 for vertical and horizontal displacements

TEST METHODS: (cont'd)

Hardness

4. ASTM C661 : 2015 Standard Test Method For Indentation Hardness Of Elastomeric-Type Sealants By Means Of A Durometer

Test Conditions:
a) 23°C and 50% relative humidity for 7 days
b) 38°C and 95% relative humidity for 7 days
c) 23°C and 50% relative humidity for 7 days

No. of determinations	:	2, 3 points per test piece

Tack-Free Time

5. ASTM C679 : 2015 Standard Test Method For Tack-Free Time Of Elastomeric Sealants

No. of determinations	:	2

Cyclic Adhesion & Cohesion

6. Adopted ASTM C719 : 2014 (2019) Standard Test Method For Adhesion And Cohesion Of Elastomeric Joint Sealants Under Cyclic Movement (Hockman Cycle)

Test Conditions:
a) 23°C and 50% relative humidity for 7 days
b) 38°C and 95% relative humidity for 7 days
c) 23°C and 50% relative humidity for 7 days
d) Immersion in distilled water at 23°C for 7 days
e) Drying in oven at 70°C for 7 days

Substrate	:	Aluminium and mortar
Test temperature	:	Room temperature
Class	:	25
No. of cycles	:	10
Crosshead speed	:	3.2 mm/hr
No. of determinations	:	3 per substrate

Effects Of Heat Ageing

7. ASTM C1246 : 2017 Standard Test Method For Effects Of Heat Ageing On Weight Loss, Cracking, And Chalking Of Elastomeric Sealants After Cure

Test Conditions:
a) 23°C and 50% relative humidity for 28 days
b) 70°C for 21 days

No. of determinations	:	3, 1 as control

TEST METHODS: (cont'd)

Effects Of Accelerated Weathering

8. Adopted ASTM C793 : 2005 (2017) Standard Test Method For Effects Of Accelerated Weathering On Elastomeric Joint Sealants

Test Conditions:
23°C and 50% relative humidity for 21 days
Test equipment : QUV Weatherometer
Test cycle : 8 hours UV exposure at 60±3°C and 4 hours condensation at 50±3°C, irradiance 0.89 W/m^2.nm (ASTM G154)
Lamp designation : Fluorescent UVA 340 mm
Exposure duration : 250 hours
No. of determinations : 3 (1 as control)
Bend test
Test equipment : Environmental chamber
Apparatus : Steel mandrel
Test condition : -26°C for 24 hours
No. of determinations : 3

Adhesion-In-Peel

9. ASTM C794 : 2018 Standard Test Method For Adhesion-In-Peel Of Elastomeric Joint Sealants

Test Conditions:
23°C and 50% relative humidity for 21 days
Substrate : Aluminium and mortar
Crosshead speed : 50 mm/min
No. of determinations : 4 per substrate

CONDITIONING:

Unless otherwise specified, all test specimens were tested at 23 ± 2°C and 50 ± 5% relative humidity.
Standard Conditions parameters: 23 ± 2°C and 50 ± 5% relative humidity.

TEST RESULTS:

Test	'Bostik A330 Fill A Gap Flex Acrylic Sealant For Sealing And Gap Filling'	HDB Specifications Sealant - Semi-Precast Construction
1. Staining And Colour Change	No staining No colour change	No visible staining on white cement mortar base
2. Extrudability	578.5ml/min	> 10ml/min
3. Rheological (Flow) Properties	Vertical displacement: 0mm sag Horizontal displacement: No deformation	Vertical displacement: < 4.8mm Horizontal displacement: no deformation
4. Indentation Hardness	test piece 1, average : 18.9 test piece 2, average : 20.0 average of 2 test pieces : 19.5	25 to 50 (traffic) 15 to 50 (non-traffic)
5. Tack-Free Time	No transfer of test specimens to the polyethylene film after 50mins	No transfer of sealant to PE film

TEST RESULTS: (cont'd)

Test	'Bostik A330 Fill A Gap Flex Acrylic Sealant For Sealing And Gap Filling'	HDB Specifications Sealant - Semi-Precast Construction
6. Adhesion & Cohesion Under Cyclic Movement, Class 25 a. Aluminium b. Mortar	No loss in bond No loss in bond	Total loss in bond and adhesion $< 9cm^2$
7. Effects Of Heat Ageing On Weight Loss, Cracking And Chalking, average	0.4% No cracking and chalking	Loss in weight < 7% No cracking and chalking
8. Effects Of Accelerated Weathering	No cracks after UV exposure and bend test	No cracks
9. Adhesion-In-Peel, average a. Aluminium b. Mortar	52.5N 51.8N cohesive failure within the sealant and no adhesive bond loss between sealant and substrate for each test piece	Peel strength >22.2N Bond loss <25%

REMARKS:

1. The test conditions for staining and colour change tests and effects of accelerated weathering test were adopted from ASTM G154 : 2016 Standard Practice For Operating Fluorescent Light Apparatus For UV Exposure Of Non-Metallic Materials.
2. The tests were specified by the client.
3. The substrates did not require primer before application of the sealant as specified by the client.
4. The class and types of substrate were specified by the client for ASTM C719 joint movement and ASTM C794 peel strength tests.
5. As specified by the client, the material identification/verification FTIR test was not required.
6. Notes in HDB specifications:
   a. The sample shall be tested for new and renewal listing.
   b. The sample for testing shall be selected by officers from HDB.

Lem Chee Meng
Testing Officer

Eddie Suwand
Senior Associate Engineer
Real Estate & Infrastructure
Mechanical Centre

PSB Singapore

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Effective 26 January 2021